

Gesture-Controlled Virtual Mouse - Comprehensive Project Report

1. Introduction

The Gesture-Controlled Virtual Mouse is an advanced human-computer interaction system that replaces the traditional computer mouse with hand gestures.

This system uses a combination of computer vision and artificial intelligence to recognize and interpret hand movements in real-time, allowing users to control the cursor, click, drag, and scroll using predefined gestures.

The rise of touch-free interfaces has accelerated research into gesture-based controls. The demand for contactless interaction

has increased, especially in applications related to accessibility, gaming, smart home automation, and virtual reality.

1.1 Background and Motivation

Traditional input devices like the mouse and keyboard have limitations, especially for users with mobility impairments. The advent of artificial intelligence,

deep learning, and computer vision has opened new possibilities for hands-free interaction with computers. Gesture control technology has gained traction in:

- **Gaming Industry:** Enhancing immersive experiences.
- **Healthcare:** Assisting physically challenged individuals.
- **Smart Homes & IoT:** Allowing users to control devices with gestures.
- **Virtual and Augmented Reality:** Improving interaction with digital environments.

1.2 Scope of the Project

This project aims to create a robust gesture-controlled system that:

- Recognizes different hand gestures accurately.
- Maps gestures to corresponding mouse functions.
- Works under different lighting conditions and backgrounds.
- Provides an alternative input method for accessibility applications.
- Uses a cost-effective approach requiring only a standard webcam.

2. Objectives

- Develop a system that tracks hand gestures to perform basic mouse functions.
- Eliminate the need for a physical mouse, improving accessibility.
- Implement real-time gesture recognition with high accuracy.
- Integrate voice commands for additional functionality.
- Optimize for different lighting conditions and hand orientations.

3. Literature Review

Gesture-based interactions have been widely explored in Human-Computer Interaction (HCI).

Research indicates that using hand gestures for controlling digital interfaces improves efficiency and reduces physical strain. Technologies such as OpenCV, MediaPipe, and Deep Learning have enabled precise hand tracking and gesture recognition.

Existing technologies include:

- **Leap Motion:** Uses infrared sensors for hand tracking.
- **Microsoft Kinect:** Depth-sensing camera for gesture recognition.
- **Wearable Gesture Devices:** Gloves with embedded sensors for motion detection.

Our project aims to provide a software-based approach using a standard webcam without additional hardware.

4. System Architecture

The system follows a modular design with the following components:

- **Hand Tracking Module** (detects hand and fingers using OpenCV and MediaPipe)
- **Gesture Recognition Module** (maps gestures to predefined mouse actions)
- **Voice Command Module** (uses speech recognition for additional commands)
- **User Interface** (integrates gesture input into a functional desktop environment)

A detailed system architecture diagram is included to show interactions between these modules.

5. Technologies Used

- **Python** for scripting and machine learning integration.
- **OpenCV** for real-time video processing.
- **MediaPipe** for hand tracking and landmark detection.
- **PyAutoGUI** for simulating mouse movements.
- **SpeechRecognition** for voice command support.
- **Flask/Web Interface** for optional UI enhancements.

6. Methodology

1. Capture real-time video feed from the webcam.
2. Detect hand landmarks using MediaPipe's hand tracking model.
3. Identify gestures based on landmark positions.
4. Map detected gestures to mouse events (clicks, drags, scrolls, etc.).
5. Process voice commands for additional functionalities.
6. Provide feedback via on-screen indicators.

7. Implementation Details

Hand Gesture Recognition

The system uses MediaPipe Hands, which provides 21 hand landmarks for detecting finger positions. Different gestures correspond to:

- Index finger up → Mouse Movement
- Thumb & Index finger touch → Left Click
- Two fingers pinch → Right Click
- Swipe Up/Down → Scroll

8. User Interface

The system runs as a background service, with an optional web-based interface to configure settings and enable/disable features.

The UI also displays real-time feedback on recognized gestures. Screenshots of the interface and interaction results are included.

9. Results & Performance

- **Gesture recognition accuracy:** ~90% under ideal lighting conditions.
- **Voice command recognition:** ~85% depending on background noise.
- **Processing Speed:** Runs in real-time at ~30 FPS on mid-range hardware.

Performance evaluation data and test cases are included to demonstrate system reliability.

10. Challenges Faced

- Hand detection accuracy varies with lighting conditions.
- Background noise affects speech recognition.
- Certain gestures may be misinterpreted if hands are partially visible.

11. Future Improvements

- **AI-based gesture learning:** Custom gestures for personalized experience.
- **Support for multi-hand interactions.**
- **Integration with AR/VR for immersive applications.**
- **Cross-platform compatibility** (Windows, macOS, Linux).

12. Case Studies and Applications

Gesture control can be applied in:

- **Gaming:** Enhancing gameplay experiences.
- **Healthcare:** Assisting people with mobility impairments.
- **Smart Homes:** Controlling IoT devices hands-free.
- **Virtual Reality (VR) & Augmented Reality (AR):** Enabling immersive interactions.

Case studies on usability tests and real-world implementations are included.

13. Conclusion

The Gesture-Controlled Virtual Mouse project successfully demonstrates an alternative to traditional input devices using computer vision and AI.

With further refinements, such technology can be widely adopted in gaming, accessibility solutions, and smart devices. This report serves as an extensive guide on the project's implementation, challenges, and future scope.

14. References

- MediaPipe Documentation (<https://developers.google.com/mediapipe>)
- OpenCV Official Docs (<https://docs.opencv.org>)
- SpeechRecognition Library (<https://pypi.org/project/SpeechRecognition>)

SOURCE CODE

```
<!DOCTYPE html>

<html>

<head>

  <title>Proton</title>

  <script type="text/javascript" src="main.js" defer></script>

  <link rel="stylesheet" type="text/css" href="css/jquery.convform.css">

</head>

<body>

  <!-- header -->

  <header>

    <div class = "header-wrapper">

      <p class="awesome">PROTON Welcomes you!</p>

      <div class = "logo">

        <img src= "images/icon.png" width="35" height="33">
```

</div>

</div>

</header>

<!-- ChatBot -->

<div class="chat_icon">

<i class="fa fa-comments" aria-hidden="true"></i>

</div>

<div class="chat_box">

<div class="conv-form-wrapper" style="height:auto;">

<div id="messages" style="overflow-y:auto;height:400px;">

</br></br>

</div>

</div>

<div id="convForm" class="convFormDynamic"
style="height:50px;position:absolute;left:10px;bottom:0px;margin:5px;">

<div class="options dragscroll"></div>

<input type="text" id="userInput" placeholder="Type Here"
class="userInputDynamic" style="margin: 7px 7.575px 7px 10px; width: 219px; height:
24px;">

<button id="userInputButton" type="button" class="submit" style="margin-
top:9px">▶</button>

</div>

</div>

```
</body>

<script type="text/javascript" src="js/main.js" defer ></script>

</html>
```

JS Code

```
//user clicked button

document.getElementById("userInputButton").addEventListener("click", getUserInput,
false);

//user pressed enter '13'

document.getElementById("userInput").addEventListener("keyup", function (event) {

    if (event.keyCode === 13) {

        //cancel the default action

        event.preventDefault();

        //process event

        getUserInput();

    }

});

eel.expose(addUserMsg);

eel.expose(addAppMsg);

function addUserMsg(msg) {

    element = document.getElementById("messages");
```

```

element.innerHTML += '<div class="message from ready rtoI">' + msg + '</div>';

element.scrollTop = element.scrollHeight - element.clientHeight - 15;

//add delay for animation to complete and then modify class to => "message from"

index = element.childElementCount - 1;

setTimeout(changeClass.bind(null, element, index, "message from"), 500);
}

function addAppMsg(msg) {

    element = document.getElementById("messages");

    element.innerHTML += '<div class="message to ready ltor">' + msg + '</div>';

    element.scrollTop = element.scrollHeight - element.clientHeight - 15;

    //add delay for animation to complete and then modify class to => "message to"

    index = element.childElementCount - 1;

    setTimeout(changeClass.bind(null, element, index, "message to"), 500);
}

function changeClass(element, index, newClass) {

    console.log(newClass + ' ' + index);

    element.children[index].className = newClass;
}

function getUserInput() {

    element = document.getElementById("userInput");

    msg = element.value;

    if (msg.length != 0) {

        element.value = ""

        eel.getUserInput(msg);

    }
}

```


PROJECT REPORT

**VIRTUALLY CONTROLLING A COMPUTER
USING HAND GESTURES AND VOICE
COMMANDS**



UNDER THE GUIDANCE OF

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GRAPHIC ERA HILL UNIVERSITY

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